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Drain hose restriction device for washing machine and washing machine

Abstract

A drain hose restriction device and a washing machine are provided, comprising: a base plate for detachably connecting with a body. The drain hose restriction device includes a rope fastening structure, comprising a first part having a mounting cavity and a second part, at least part of the second part being disposed inside the mounting cavity; an elastic member, disposed inside the mounting cavity, with one end against or connected to a wall of the cavity of the mounting cavity, and the other end being pressed against or connected to the second part, being always in a compressed state; a limiting rope, one end of which is threaded through the first part and the second part and is connected to the base plate,, the limiting rope positioned at one side of the first part back away from the base plate forms a limiting ring.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application claims priority to and benefits of Chinese Patent Application No. 2023100563852 filed with China National Intellectual Property Administration on Jan. 19, 2023 and entitled “drain hose restriction device for washing machine and washing machine”, the entire contents of each of which are incorporated herein by reference for all purposes. No new matter has been introduced.
FIELD
(2) The present disclosure relates to the technical field of washing machines, and specifically to a drain hose restriction device and a washing machine.
BACKGROUND
(3) As the popularity of cleaning equipment such as washing machines gradually increases, how to enhance the storage effect and user experience is a key concern of various manufacturers.
(4) In the related art, as shown in FIGS. **1** and **2**, one end of the drain hose **220′** is connected to a drainage chamber of the washing machine **200′**, and the other end of the drain hose **220′** is suspended from the outer wall of the washing machine body **210′** by a restriction structure **230′**. The restriction structure **230′** includes a first slot **231′**, a second slot **232′**, and a suspension hook **233′**. Specifically, the first slot **231′** is disposed on the outer wall of the machine body **210′**, and the

second slot **232′** is usually disposed at the other end (the end provided with the water outlet) of the drain **220′**. The suspension hook **233′** is set in the second slot **232′**, and the suspension hook **233′** and the first slot **231′** cooperate with each other to realize the suspension limitation of the drain hose **220′**.

(5) In the related art, as shown in FIGS. **1** and **2**, one end of the drain hose **220′** is connected to the drainage cavity of the washing machine **200′**, and the other end of the drain hose **220′** is suspended from the machine body **210′** of the washing machine **200′** through the limiting structure **230′**. outer wall. The limiting structure **230′** includes a first groove **231′**, a second groove **232′** and a suspension hook **233′**. Specifically, the first groove **231′** is provided on the outer wall of the machine body **210′**, and the second groove **232′** is usually provided at the other end of the drain hose **220′** (the end with the water outlet). The suspension hook **233′** is set in the second groove **232′**, and the suspension hook **233′** cooperates with the first groove **231′** to realize the hanging limit of the drain hose **220′**.

(6) This design method has the following limitations: the suspension hook **233′** May break (easy to break at A in FIG. **2**), and after breaking, it cannot cooperate with the first slot **231′**, and at this time, the limiting structure **230′** no longer has a limiting function; the limiting structure **230′** of the second slot **232′** and the suspension hook **233′** are usually provided at the end of the drain hose **220′** provided with a water outlet. Users may replace the drain hose **220′** with different lengths according to their needs. In the case that the drain hose **220′** is too short, the suspension hook **233′** May not be able to reach the first slot **231′**; in the case that the drain hose **220′** is too long, the storage effect of the drain hose **220′** is not ideal when the suspension hook **233′** and the first slot **231′** are fitted.

(7) Further, the Chinese invention patent with authorization number CN205012064U discloses a hanging device for a washing machine drain hose, including lugs, hooks, a drain hose, and an outer casing. The invention patent realizes the suspension limitation of the drain hose through the mutual cooperation of the hook and the lugs, but still fails to solve the problem that the hook is easy to break.

SUMMARY

(8) In order to solve or ameliorate at least one of the above technical problems, it is an object of the present disclosure to provide a drain hose restriction device.

(9) Another object of the present disclosure is to provide a washing machine having a drain hose restriction device as described above.

(10) To achieve the above purposes, the first aspect of the present disclosure provides a drain hose restriction device. It includes: a base plate, detachably connected to the machine body; a buckle structure, comprising a first part with an installation cavity and a second part at least partially located within the installation cavity; an elastic member, positioned within the installation cavity and remaining in a compressed state, with one end abutting or connected to the cavity wall of the installation cavity, and with the other end abutting or connected to the second part; a limiting rope, with an end threaded through the first part and the second part, connected to the base plate, and with the other end threaded through the first part and the second part, also connected to the base plate, the limiting rope positioned at one side of the first part back away from the base plate forming a limiting ring which is fitted onto the drain hose to limit its position, the first part and the second part being capable of moving relative to the base plate to change the size of the limiting ring.

(11) According to the technical solution of the drain hose restriction device proposed by the present disclosure, the device does not require a suspension hook, and the drain hose can be limited by the limiting ring formed by the limiting rope. Furthermore, the base plate of the drain hose restriction device is detachably connected to the machine body, facilitating the installation of the device at any desired position on the machine body. This allows the limiting ring formed by the limiting rope to effectively limit the position of the drain pipe, regardless of its length, resulting in improved

storage efficiency.
(12) Specifically, the drain hose restriction device includes a base plate and a rope buckle structure. Wherein, the base plate is used for detachably connecting with the machine body of the washing machine. It can be understood that the drain hose restriction device is detachably connected to the machine body of the machine, allowing users to install the device at any position of the machine body. Optionally, the base plate and the outer wall of the machine body are detachably connected by means of adhesive bonding, which is easy for users to operate. Alternatively, the base plate and the machine body are detachably connected using a connecting member, which is easy for users to operate. Optionally, the connecting member is a screw or a self-tapping screw. The connecting member is threaded through the base plate and the machine body. The connection element is threadedly connected to the base plate, and the connecting member is threadedly connected to the machine body.
(13) Further, the rope buckle structure includes a first part, a second part, an elastic member, and a limiting rope. Specifically, the first part is disposed on the side of the base plate back away from the machine body. The first part has a mounting cavity. Further, the second part is disposed on the side of the base plate back away from the machine body. At least a portion of the second part is disposed within the mounting cavity. In other words, at least a portion of the second part is disposed in the mounting cavity of the first part. It should be noted that the first portion and the second portion may be of any shape according to practical needs. Further, the elastic member is disposed in the mounting cavity. One end of the elastic member is pressed against or connected to the wall of the mounting cavity. The other end of the elastic member is pressed against or connected to the second part. Further, one end of the limit rope threaded through and connected to both the first part and the second part, as well as the base plate. The other end of the limit rope is also threaded through and connected to both the first part and the second part, as well as the base plate. The elastic member is always in a compressed state. The second part always tends to be detached from the mounting cavity of the first part under the action of the elastic force of the elastic member. Therefore, the limiting rope threaded through the first part and the second part are subjected to significant friction. Without external force, it is difficult for the limiting rope to change their positions with respect to the first part or the second part. When users press down on the second part (applying an external force), the elastic member inside the installation cavity is further compressed. At this time, the friction on the limiting rope which is threaded through the first part and the second part decreases, allowing the first part and the second part to slide on the limiting rope, thus changing the positions of the limiting rope with respect to the first part and the second part.
(14) Further, the limiting rope positioned at one side of the first part back away from the base plate forms a limiting ring. The first part and the second part are able to move relative to the base plate to change the size of the limiting ring. By adjusting the size of the limiting ring, it can accommodate a wider range of drain hoses, making it highly versatile. Furthermore, the limiting ring is fitted onto the drain hose to provide limitation for the drain hose.
(15) In the technical solution described by the present disclosure, the drain hose restriction device does not require a suspension hook, instead utilizes a limiting rope to form a limiting ring for restricting the drain hose. Furthermore, the base plate of the drain hose restriction device is detachably connected to the machine body, allowing for convenient installation at any desired position. This design enables the limiting ring formed by the limiting rope to effectively restrict the drain pipe at various positions, regardless of its length, thereby enhancing the storage effect.
(16) Furthermore, the above technical solution provided by the present disclosure can also have the following additional technical features:
(17) In the above technical solution, the first part is provided with two first holes that are in communication with the installation cavity, and the second part is provided with two second holes, and each end of the limiting rope is threaded through a corresponding first hole as well as the

second hole.
(18) In this technical solution, by pre-setting a first hole on the first part and a second hole on the second part, it is convenient to quickly thread the limiting rope through the first part and the second part. The second part always has a tendency to be detached from the mounting cavity of the first part under the action of the spring force. The centerline of the first hole and the centerline of the second hole have a tendency to move away from each other, so that the limiting rope threaded through the first hole and the second hole are subjected to a large friction force. In the absence of an external force, it is difficult for the limit rope to change its position relative to the first part or the second part. Users presses down on the second part (applying an external force) to further compress the elastic member in the mounting cavity, at which time the centerline of the first hole and the centerline of the second hole are close to each other, so that the friction applied to the limit rope is reduced, and the first part as well as the second part can be slid on the limit rope.
(19) In the above technical solution, the elastic member is a spring; or the elastic member is a leaf spring; or the elastic member is a rubber block.
(20) In this technical solution, by setting the elastic member as a spring or a leaf spring or a rubber block, and the elastic member is always in a compressed state, the second part always has a tendency to be detached from the mounting cavity of the first part under the action of the spring's elastic force, and therefore the limit rope threaded through the first part and the second part are subjected to a great friction. It is difficult for the limit rope to have a change of relative positions with the first part or the second part in the absence of the action of an external force. Users presses down on the second part (applying an external force) to further compress the elastic member in the mounting cavity, at which time the friction of the limit rope threaded through the first part and the second part decreases, and the first part as well as the second part can slide on the limit rope to change their positions relative to the limit rope.
(21) In the above technical solution, the base plate is adhered to the outer wall of the machine body.
(22) In this technical solution, the base plate and the outer wall of the machine body are detachably connected by means of adhesion, which is easy for users to operate. In the technical solution described by the present disclosure, the added drain hose restriction device does not need to change the original structure of the washing machine, and is highly versatile. Optionally, the drain hose restriction device is an independent structure. One of the sides of the base plate is provided with an adhesive layer and a protective layer. When it is necessary to use the drain hose restriction device, users can tear off the protective layer and directly press the adhesive layer against the machine body.
(23) In the above technical solution, the base plate and the machine body are detachably connected by means of a connecting member.
(24) In this technical solution, by providing the connecting member, it is possible to realize a detachable connection between the base plate and the machine body, which is easy to be operated by users. It is worth stating that the number of the connecting member is at least one, i.e., the connecting member may be one, two, or more than one. Considering factors such as convenience of operation, connection strength between the base plate and the machine body in the connected state, cost, and other factors, the connecting member is flexibly set according to specific requirements and practical needs. Optionally, the connecting member is a screw. Optionally, the connecting member is a self-tapping screw.
(25) In the above technical solution, the connecting member is a self-tapping screw, which is threaded with the base plate and the machine body.
(26) In this technical solution, the connectors are designed as self-tapping screws, making it convenient for users to install and remove the drain hose restriction device. Additionally, the drain hose restriction device can be installed at any position of the machine body.
(27) In the above technical solution, the drain hose restriction device further comprises at least two fixing portions for connecting the base plate to one end of the limiting rope and connecting the base

plate to the other end of the limiting rope.
(28) In this technical solution, the drain hose restriction device further comprises at least two fixing portions. Specifically, the at least one fixing portion is for connecting the base plate to one of the ends of the limiting rope. The at least one fixing portion is for connecting the base plate with the other end of the limiting rope. By providing the fixing portion, it is possible to connect an end of the limit rope to the base plate. Each of the two ends of the limit rope is connected to the base plate by the at least one fixing portion.
(29) In the above technical solution, the rope buckle structure further comprises a pulling portion disposed in the limit ring.
(30) In this technical solution, the rope buckle structure further comprises a pulling portion. Specifically, the pulling portion is located on the limit ring. By providing the pulling portion, users can easily pull the limiting rope, enhancing user experience. Optionally, the pulling portion can be of any shape.
(31) In the above-described technical solution, the rope buckle structure further comprises a pressing portion, which is disposed at an end of the second portion that is outside the mounting cavity.
(32) In this technical solution, the rope buckle structure further comprises a pressing portion. Specifically, the pressing portion is provided at an end of the second part that is outside the mounting cavity. By providing the pressing portion, it is convenient for users to press down on the second part, improving users experience. Optionally, the pressing portion may be a slot structure.
(33) A second aspect of the present disclosure provides a washing machine comprising: a machine body; a drain hose, which is connected to the machine body; and a drain hose restriction device in any of the above technical solutions, wherein a base plate of the drain hose restriction device is detachably connected to the machine body, and a limiting ring of the drain hose restriction device is fitted onto the drain hose.
(34) According to a technical solution of the washing machine of the present disclosure, the washing machine comprises a body, a drain hose, and a drain hose restriction device in any of the above technical solutions. Specifically, a base plate of the drain hose restriction device is detachably connected to the machine body. It can be understood that the drain hose restriction device is detachably connected to the machine body, making it convenient for users to install the drain hose restriction device at any position of the machine body. Optionally, the base plate and the outer wall of the machine body are detachably connected by means of adhesion, which is easy for users to operate. Optionally, the base plate and the machine body are detachably connected by means of a connecting member, which is easy for users to operate. Optionally, the connecting member is a screw or a self-tapping screw. The connecting member is threaded through the base plate and the machine body. The connecting member is threadedly connected to the base plate and the machine body. Further, a limiting ring of the drain hose restriction device is fitted onto the drain hose to restrict the drain hose.
(35) Wherein, since the washing machine comprises any of the drain hose restriction devices in the first aspect described above, it has the beneficial effect of any of the technical solutions described above, which will not be repeated herein.
(36) Additional aspects and advantages of the technical embodiments of the present disclosure will become apparent in the following descriptive portions or through the practice of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) What are described above and/or additional aspects and advantages of the present disclosure will become obvious and comprehensible from the following description of embodiments in

conjunction with the accompanying drawings.
(2) FIG. **1** illustrates a schematic diagram of a washing machine in the related art;
(3) FIG. **2** illustrates a schematic view of a drain hose in the related art;
(4) FIG. **3** illustrates a schematic view of a drain hose restriction device according to an embodiment of the present disclosure;
(5) FIG. **4** illustrates a schematic view of a washing machine according to an embodiment of the present disclosure.
(6) Wherein, the correspondence between the accompanying markings and the names of the parts in FIGS. **1** and **2** is: **200′**: washing machine; **210′**: machine body; **220′**: drain hose; **230′**: restriction structure; **231′**: first slot; **232′**: second slot; **233′**: suspension hook.
(7) The correspondence between the accompanying markings and the names of the parts in FIGS. **3** and **4** is: **100**: drain hose restriction device; **110**: base plate; **120**: rope buckle structure; **121**: first part; **1211**: mounting cavity; **1212**: first hole; **122**: second part; **1221**: second hole; **123**: elastic member; **124**: limiting rope; **1241**: limiting ring; **125**: pulling portion; **126**: pressing portion; **140**: fixing portion; **200**: washing machine; **210**: machine body; **220**: drain hose.
DETAILED DESCRIPTION OF THE DISCLOSURE
(8) To make the above objectives, features and advantages of the present disclosure more obvious and comprehensible, the present disclosure will be described in detail below with reference to accompanying drawings and specific embodiments. It should be noted that embodiments in the present disclosure and features in the embodiments can be combined with one another if there is no conflict.
(9) Many specific details are set forth in the following description to facilitate full understanding of the present disclosure, but the present disclosure can further be implemented in other ways different from those described herein, and therefore, the protection scope of the present disclosure is not limited by the specific embodiments disclosed below.
(10) A drain hose restriction device **100** and a washing machine **200** provided according to some embodiments of the present disclosure are described below with reference to FIGS. **3** and **4**.
(11) In an embodiment according to the present disclosure, as shown in FIG. **3**, the drain hose restriction device **100** comprises a base plate **110** and a rope buckle structure **120**. wherein the base plate **110** is used for detachably connecting with the body **210** of the washing machine **200**. It can be understood that the drain hose restriction device **100** is detachably connected to the machine body **210** to facilitate users to install the drain hose restriction device **100** at any position of the machine body **210** (the drain hose restriction device **100** is provided at any position convenient for storing the drain hose **220**). Optionally, the base plate **110** and the outer wall of the machine body **210** are detachably connected by means of adhesion, which is easy for users to operate. Optionally, the base plate **110** and the machine body **210** are detachably connected by the connecting member **130**, which is easy to be operated by users. Optionally, the connecting member **130** is a screw or a self-tapping screw. The connecting member **130** is disposed on the base plate **110** and the machine body **210**. the connecting member **130** is threadedly connected to the base plate **110** and the machine body **210**.
(12) Further, as shown in FIG. **3**, the rope buckle structure **120** includes a first portion **121**, a second portion **122**, an elastic member **123**, and a limiting rope **124**. specifically, the first portion **121** is disposed on the side of the base plate **110** that is backed away from the machine body **210**. The first portion **121** has a mounting cavity **1211**. further, the second portion **122** is disposed on the side of the base plate **110** back from the machine body **210**. At least a portion of the second portion **122** is disposed within the mounting cavity **1211**. In other words, at least a portion of the second portion **122** is provided within the mounting cavity **1211** of the first portion **121**. It is worth stating that the first portion **121** and the second portion **122** may be of any shape according to practical needs. Further, the elastic member **123** is disposed within the mounting cavity **1211**. One end of the elastic member **123** is pressed against or connected to a cavity wall of the mounting cavity **1211**.

The other end of the elastic member **123** is pressed against or connected to the second portion **122**. Further, one end of the limit rope **124** is threaded through the first portion **121** and the second portion **122** and is connected to the base plate **110**. The other end of the limit rope **124** is threaded between the first portion **121** and the second portion **122** and connected to the base plate **110**. The elastic member **123** is always in a compressed state. The second portion **122** always tends to detach from the mounting cavity **1211** of the first portion **121** under the action of the spring force. Therefore, the limit rope **124** threaded through the first portion **121** and the second portion **122** is subjected to a great friction, and it is very difficult for the limit rope **124** to change its relative position with the first portion **121** or the second portion **122** without the action of an external force. Users presses down on the second part **122** (applying an external force) to further compress the elastic member **123** in the mounting cavity **1211**, at which time the frictional force of the limit rope **124** threaded through the first part **121** and the second part **122** is reduced, and the first part **121** and the second part **122** can slide along the limit rope **124** to change the relative position of the first part **121** or the second part **122** to the limit rope **124**.
(13) Further, the limiting rope **124** is disposed on a side of the first part **121** back from the base plate **110** to form the limiting ring **1241**. the first part **121** and the second part **122** are capable of moving relative to the base plate **110** to change the size of the limiting ring **1241**. By changing the size of the limit ring **1241**, it can accommodate a wider range of sizes for the drain hose **220**, thus enhancing its versatility. Furthermore, the limit ring **1241** is fitted onto the drain hose **220** to restrict its movement.
(14) In the technical solution described by the present disclosure, the drain hose restriction device **100** does not require a suspension hook, instead utilizes a limiting rope **124** to form a limiting ring **1241** for restricting the drain hose **220**. In addition, the base plate **110** of the drain hose restriction device **100** is detachably connected to the machine body **210**, allowing for convenient installation of the device **100** at any desired position of the machine body **210**. This design enables the limiting ring **1241** formed by the limiting rope **124** to effectively restrict the drain hose **220** at various positions, regardless of its length, thereby enhancing the storage effect.
(15) In an embodiment according to the present disclosure, as shown in FIG. **3**, the first part **121** is provided with two first holes **1212** which are connected to the mounting cavity **1211**, and the second part **122** is provided with two second holes **1221**, and each of the two ends of the limiting rope **124** is threaded through the corresponding one first hole **1212** and one second hole **1221**. By presetting the first hole **1212** on the first part **121** and the second hole **1221** on the second part **122**, it is convenient to quickly pass the limiting rope **124** through the first part **121** and the second part **122**. under the action of spring force, the second part **122** always tends to detach from the mounting cavity **1211** of the first part **121**. The centerline of the first hole **1212** and the centerline of the second hole **1221** tend to move away from each other, resulting in significant friction on the limit rope **124** threaded through the first hole **1212** and the second hole **1221**. Without external force, it is difficult for the limit rope **124** to change its relative position with the first part **121** or the second part **122**. When the user presses down on the second part **122** (applying external force), the elastic component **123** inside the installation chamber **1211** further compresses. At this time, the centerline of the first hole **1212** and the centerline of the second hole **1221** approach each other, reducing the friction on the limit rope **124**. As a result, the first part **121** and the second part **122** can slide along the limit rope **124**.
(16) In an embodiment according to the present disclosure, the elastic member **123** is a spring; or the elastic member **123** is a leaf spring; or the elastic member **123** is a rubber block. By setting the elastic component **123** as a spring, elastic sheet, or rubber block and keeping it in a compressed state, the second part **122** always tends to detach from the installation chamber **1211** of the first part **121** under the action of the spring force. Therefore, the limit rope **124** threaded through the first part **121** and the second part **122** will experience significant friction. Without external force, it is difficult for the limit rope **124** to change its relative position with the first part **121** or the second

part **122**. When the user presses down on the second part **122** (applying external force), the elastic component **123** inside the installation chamber **1211** further compresses. As a result, the friction on the limit rope **124** threaded through the first part **121** and the second part **122** decreases, allowing the first part **121** and the second part **122** to slide along the limit rope **124** and change their relative positions with the limit rope **124**.
(17) In an embodiment according to the present disclosure, the base plate **110** and the outer wall of the machine body **210** are detachably connected by means of adhesion, which is easy to be operated by users. In the technical solution limited by the present disclosure, the newly added drain hose restriction device **100** does not require any modification to the existing structure of the washing machine **200**, making it highly versatile. Optionally, the drain hose restriction device **100** is a separate structure. One side of the base plate **110** is provided with an adhesive layer a protective layer. When it is necessary to use the drain hose restriction device **100**, users can tear off the protective layer and directly place the adhesive layer against the machine body **210**.
(18) In an embodiment according to the present disclosure, the base plate **110** and the machine body **210** are detachably connected by the connecting member **130**. By providing the connecting member **130**, the detachable connection between the base plate **110** and the machine body **210** can be realized, which is easy to be operated by users. It is worth stating that the number of the connecting members **130** is at least one, i.e., the connecting members **130** may be one, two, or more than one. Considering factors such as convenience of operation, connection strength between the base plate and the machine body in the connected state, cost, and other factors, the connecting member **130** is flexibly set according to specific requirements and practical needs. Optionally, the connecting member **130** is a screw. Optionally, the connecting member **130** is a self-tapping screw.
(19) Further, the connecting member **130** is a self-tapping screw which is threadedly connected to the base plate **110** and the machine body **210**. By setting the connecting member **130** as a self-tapping screw, it is convenient for a user to install and remove the drain hose restriction device **100**, and the drain hose restriction device **100** can be installed at any position of the machine body **210**.
(20) In an embodiment according to the present disclosure, as shown in FIG. **3**, the drain hose restriction device **100** further comprises at least two fixing portions **140**. specifically, the at least one fixing portion **140** is for connecting the base plate **110** to one end of the limiting rope **124**. The at least one fixing portion **140** is for connecting the base plate **110** with the other end of the limit rope **124**. By providing the fixing portion **140**, it is possible to connect an end of the limit rope **124** to the base plate **110**. Each of the two ends of the limit rope **124** is connected to the base plate **110** by the at least one fixing portion **140**.
(21) In an embodiment according to the present disclosure, as shown in FIG. **3**, the rope buckle structure **120** further comprises a pulling portion **125**. Specifically, the pulling portion **125** is provided at the limit ring **1241**. By providing the pulling portion **125**, users can easily pull the limit rope **124**, enhancing user experience. Optionally, the pulling and portion **125** can be of any shape.
(22) In an embodiment according to the present disclosure, as shown in FIG. **3**, the rope buckle structure **120** further comprises a pressing portion **126**. specifically, the pressing portion **126** is provided at an end of the second portion **122** that is outside the mounting cavity **1211**. By providing the pressing portion **126**, users can easily press down on the second portion **122**, enhancing user experience. Optionally, the pressing portion **126** may be a slot structure.
(23) In an embodiment according to the present disclosure, the limiting ring **1241** of the rope buckle structure **120** may be adjustable in caliber to facilitate limiting of the drain hose **220**. Optionally, the limiting rope **124** may be disposed in a second slot of the drain hose **220** to limit the drain hose **220**. It is worth stating that the limiting rope **124** can restrict any position of the drain hose **220** (regardless of the length of the drain hose **220**) for better storage.
(24) In an embodiment according to the present disclosure, as shown in FIG. **4**, the washing machine **200** comprises a machine body **210**, a drain hose **220**, and a drain hose restriction device **100** in any of the above embodiments. Specifically, the base plate **110** of the drain hose restriction

device **100** is detachably connected to the machine body **210**. It is to be understood that the drain hose restriction device **100** is detachably connected to the machine body **210** to facilitate a user to install the drain hose restriction device **100** at any position of the machine body **210**. Optionally, the base plate **110** is detachably connected to the outer wall of the machine body **210** by means of adhesion, which is easy for users to operate. Optionally, the base plate **110** and the machine body **210** are detachably connected by means of a connecting member **130**, which is easy to be operated by users. Optionally, the connecting member **130** is a screw or a self-tapping screw. The connecting member **130** is threaded through the base plate **110** and the machine body **210**. the connecting member **130** is threadedly connected to the base plate **110** and the machine body **210**. Further, the limiting ring **1241** of the drain hose restriction device **100** is fitted onto the drain hose **220** to restrict the drain hose **220**.

(25) According to the embodiment of the drain hose restriction device and the washing machine of the present disclosure, the drain hose restriction device does not require a suspension hook, and the drain hose can be limited by the limiting ring formed by the limiting rope. In addition, the base plate of the drain hose restriction device is detachably connected to the machine body, which facilitates the installation of the drain hose restriction device at any position of the machine body, so that the limiting ring formed by the limiting rope can limit the drain hose at any position of the drain hose (regardless of the length of the drain hose), and the storage effect is enhanced.

(26) In the present disclosure, the terms “first”, “second”, “third” are used for descriptive purposes only and are not to be construed as indicating or implying relative importance; the term “more than one” refers to two or more, unless otherwise expressly limited. “plurality” refers to two or more, unless otherwise expressly limited. The terms “mounted”, “connected”, “linked”, “attached”, “secured”, etc. are to be understood in a broad sense, e.g. The term “connection” May be a fixed connection, a removable connection, or a connection in one piece; the term “connection” May be a direct connection or an indirect connection through an intermediate medium. For those of ordinary skill in the art, the specific meaning of the above terms in the present disclosure may be understood according to the specific circumstances.

(27) In the description of the present disclosure, it is to be understood that the terms “up”, “down”, “left”, “right”, “front”, “back”, etc. indicate orientation or positional relationships based on those shown in the accompanying drawings, and are intended only to facilitate and simplify the description of the present disclosure, and do not indicate or imply that the device or unit referred to must be oriented, constructed and operated in a particular direction, or in a particular orientation. orientation to be constructed and operated and, therefore, are not to be construed as a limitation of the present disclosure.

(28) In the description of the present disclosure, the terms “an embodiment”, “some embodiments”, “specific embodiment”, etc. indicate that specific features, structures, materials or characteristics described in conjunction with the embodiment or illustrative description are included in at least one embodiment or instance of the present disclosure. In the present disclosure, the schematic description of the above terms does not necessarily refer to the same embodiment or instance. Moreover, the specific features, structures, materials or characteristics described can be combined in a suitable way in any one or more embodiments or instances.

(29) What are described above are merely some embodiments of the present disclosure and are not intended to limit the present disclosure, and various changes and modifications can be made by those skilled in the art. Any modifications, equivalent substitutions, improvements, etc. made within the spirit and principle of the present disclosure should fall within the protection scope of the present disclosure.

Claims

1. A washing machine, comprising: a machine body (**210**); a drain hose (**220**), connected to said body (**210**); a drain hose restriction device (**100**) comprising: a base plate (**110**), for removable attached to said machine body (**210**); a rope buckle structure (**120**), including first section (**121**) with a mounting cavity (**1211**) and second portion (**122**), at least part of said second portion (**122**) being disposed within said mounting cavity (**1211**); an elastic member (**123**), provided within said mounting cavity (**1211**), one end of said elastic member (**123**) being pressed against or connected to a wall of said mounting cavity (**1211**), and the other end of said elastic member (**123**) being pressed against or connected to said second portion (**122**), with said elastic member (**123**) being always in a compressed state; a limiting rope (**124**), one end of said limiting rope (**124**) is threaded through said first part (**121**) and said second part (**122**) and connected to said base plate (**110**), the other end of said limiting rope (**124**) is threaded through said first part (**121**) and said second part (**122**) and is connected to said base plate (**110**), said limiting rope (**124**) positioned at the side of said first part (**121**) back away from said base plate (**110**) forming a limiting ring (**1241**), said first part (**121**) and said second part (**122**) being able to move relative to said base plate (**110**) to change the size of said limiting ring (**1241**), and said limiting ring (**1241**) being fitted onto a drain hose (**220**) to limit said drain hose (**220**), wherein the base plate (**110**) of said drain hose restriction device is detachably connected to said body (**210**), and said limiting ring (**1241**) of said drain hose restriction device is fitted onto said drain hose (**220**).
2. The washing machine according to claim 1, wherein: said first part (**121**) is provided with two first holes (**1212**), said first holes (**1212**) being in communication with said mounting cavity (**1211**), and said second part (**122**) is provided with two second holes (**1221**), and each of the two ends of said limiting rope (**124**) is threaded through the corresponding one first hole (**1212**) and one second hole (**1221**).
3. The washing machine according to claim 1, wherein: said elastic member (**123**) is a spring.